

Further candidate Management Procedures towards a Scoping MP for South African sardine

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This document compares a further set of candidate Management Procedures under a reduced set of performance statistics to assist in selecting a Scoping MP for providing TAC/B recommendations for South African sardine in the short term. Results are given for the Operating Model which assumes a Beverton Holt stock recruitment relationship.

Keywords: management procedure, management strategy evaluation, performance statistics, sardine, South Africa

Introduction

A further set of candidate Management Procedures for the sardine-only Scoping MP are considered in this document. The Management Strategy Evaluation with Operating Models have been described elsewhere (e.g. de Moor 2025a,b). Table 1 and Figures 1-2 give the reduced set of performance statistics requested by the Technical Task Group of the Small Pelagic Scientific Working Group. These performance statistics include:

$p(SSB_{j,y} < RiskT^j)$ – Probability of spawner biomass being below the risk threshold (limit reference point) for stock j

$\bar{C}$ – Average catch over the projection period

$\overline{TAC}$ – Average catch over the projection period

$p(B_{c,y} < 0.7B_{c,y}^{NoC})$ – Probability that biomass off coast c under the CMP is less than 70% of that under CMP0 during the last 15 years of the projection period.

$p(B_{c,y} > 0.85B_{c,y}^{NoC})$ – Probability that biomass off coast c under the CMP is greater than 85% of that under CMP0 during the last 15 years of the projection period.

$MedB_{c,2044}^{CMP}/MedB_{c,2044}^{NoC}$ – Median biomass off coast c at the end of the projection period under the CMP : Median biomass off coast c at the end of the projection period under CMP0.¹

$MedB_{j,2044}^{CMP}/MedB_{j,2044}^{NoC}$ – Median biomass of stock j at the end of the projection period under the CMP : Median biomass of stock j at the end of the projection period under CMP0.

$MedSSB_{c,2044}^{CMP}/MedSSB_{c,2044}^{NoC}$ – Median spawner biomass off coast c at the end of the projection period under the CMP : Median biomass off coast c at the end of the projection period under CMP0.

$MedSSB_{j,2044}^{CMP}/MedSSB_{j,2044}^{NoC}$ – Median spawner biomass of stock j at the end of the projection period under the CMP : Median biomass of stock j at the end of the projection period under CMP0.

References

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de Moor CL. 2025b. Comparing an initial set of candidate Management Procedures for the Scoping MP for South African sardine. DFFE: Branch Fisheries Document FISHERIES/2025/JUL/SWG-PEL/41.

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¹ Note, this is not the same as the median in Figures 1-2. Figures 1-2 show the medians and percentiles of the biomass off coast c under a CMP to that under a no future catch scenario.

Table 1. Performance statistics for the candidate MPs tested in this document for the sardine-only Scoping MP. All TAB/Cs are given in 1000t. In all CMPs $TAB_{big}^S = 0.0332 \times B_{w,y-1}^{obs}$.

In the no small bycatch 'b' CMPs, $TAB_{y,small}^S = TAB_{y,anc}^S = 0$. In the main CMPs, $TAB_{y,small}^S = 0.00375 \times B_{w,y-1}^{obs}$ and $TAB_{y,anc}^S = 0.0072 \times B_{w,y-1}^{obs}$. In the higher bycatch with anchovy 'h' CMPs, $TAB_{y,small}^S = 0.00375 \times B_{w,y-1}^{obs}$ and $TAB_{y,anc}^S = 0.0145 \times B_{w,y-1}^{obs}$. The proportions given in the first two columns are the proportion of $B_{c,y-1}^{obs}$ used to set $TAC_{c,y}^S$.

The OMs are S_{BH} and S_0 with $F_{c,y,q,a}^{j,By} < 0.70$ denoted S_0^* below. Where appropriate, medians [95% probability intervals] are given.

	$TAC_{w,y}^S$	$TAC_{c,y}^S$	$p(SSB_{WTS,y} < RiskT_{WTS}^S)$	$p(SSB_{CTS,y} < RiskT_{CTS}^S)$	$\bar{C}$	$\bar{TAC}$	$p(B_{w,y} < 0.7B_{w,y}^{noc})$	$p(B_{w,y} > 0.85B_{w,y}^{noc})$	$p(B_{s,y} < 0.7B_{s,y}^{noc})$	$p(B_{s,y} > 0.85B_{s,y}^{noc})$	$\frac{MedB_{w,2044}^{CMP}}{MedB_{w,2044}^{NOC}}$	$\frac{MedB_{s,2044}^{CMP}}{MedB_{s,2044}^{NOC}}$	$\frac{MedB_{CTS,2044}^{CMP}}{MedB_{CTS,2044}^{NOC}}$	$\frac{MedB_{WTS,2044}^{CMP}}{MedB_{WTS,2044}^{NOC}}$	$\frac{MedSSB_{CTS,2044}^{CMP}}{MedSSB_{CTS,2044}^{NOC}}$	$\frac{MedSSB_{WTS,2044}^{CMP}}{MedSSB_{WTS,2044}^{NOC}}$
			S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}	S_{BH}
CMP0	0	0	0.216	0.021	0	0	0	0	0	0	1	1	1	1	1	1
CMP1b	0.0418	0.075	0.262	0.033	57 [19,277]	43 [13,241]	0.02	0.70	0.25	0.29	0.88	0.76	0.85	0.82	0.85	0.80
CMP2b	0.0518	0.085	0.270	0.035	63 [21,312]	50 [15,275]	0.03	0.60	0.32	0.23	0.86	0.74	0.83	0.79	0.84	0.77
CMP3b	0.0618	0.095	0.275	0.037	69 [23,346]	56 [17,308]	0.05	0.52	0.39	0.19	0.85	0.72	0.81	0.76	0.82	0.75
CMP4b	0.0718	0.105	0.282	0.039	75 [25,378]	62 [19,341]	0.08	0.44	0.45	0.15	0.83	0.70	0.79	0.73	0.81	0.73
CMP5b	0.0818	0.115	0.290	0.042	81 [27,409]	68 [21,373]	0.11	0.37	0.51	0.12	0.81	0.66	0.77	0.71	0.79	0.71
CMP6b	0.0918	0.125	0.296	0.045	86 [29,436]	74 [23,404]	0.15	0.31	0.56	0.10	0.79	0.64	0.76	0.69	0.78	0.68
CMP7b	0.1018	0.135	0.302	0.047	91 [31,463]	79 [25,436]	0.19	0.26	0.61	0.08	0.78	0.61	0.74	0.66	0.76	0.65
CMP1	0.03085	0.075	0.295	0.046	48 [16,261]	36 [10,229]	0.17	0.30	0.50	0.17	0.78	0.67	0.77	0.70	0.76	0.71
CMP2	0.04085	0.085	0.302	0.049	54 [17,294]	42 [12,262]	0.21	0.25	0.55	0.14	0.76	0.65	0.74	0.67	0.74	0.69
	0.04085	0.101	0.313	0.050	59 [19,331]	47 [13,301]	0.24	0.23	0.61	0.10	0.74	0.61	0.74	0.61	0.74	0.65
	0	0.101	0.309	0.038	44 [11,295]	31 [3,266]	0.12	0.42	0.59	0.12	0.80	0.62	0.82	0.65	0.81	0.67
CMP3	0.05085	0.095	0.310	0.052	60 [19,326]	48 [14,295]	0.25	0.21	0.59	0.11	0.74	0.62	0.73	0.65	0.73	0.67
	0.05085	0	0.254	0.052	29 [13,80]	18 [8,49]	0.15	0.34	0.16	0.57	0.80	0.85	0.72	0.87	0.73	0.86
CMP4	0.06085	0.105	0.317	0.055	65 [21,357]	53 [16,326]	0.30	0.18	0.63	0.09	0.73	0.60	0.71	0.63	0.72	0.64
CMP5	0.07085	0.115	0.323	0.058	71 [23,388]	59 [18,357]	0.34	0.15	0.66	0.08	0.71	0.58	0.70	0.61	0.70	0.62
CMP6	0.08085	0.125	0.331	0.061	75 [25,416]	64 [20,388]	0.39	0.13	0.70	0.06	0.70	0.55	0.69	0.59	0.69	0.59
CMP7	0.09085	0.135	0.338	0.064	80 [27,444]	69 [21,419]	0.43	0.11	0.73	0.05	0.69	0.53	0.68	0.57	0.67	0.57
	0.05085	0.085	0.303	0.052	57 [19,300]	45 [13,268]	0.24	0.22	0.55	0.13	0.75	0.64	0.73	0.67	0.73	0.69
	0.05085	0.101	0.313	0.052	61 [20,338]	49 [14,310]	0.26	0.20	0.61	0.10	0.74	0.62	0.73	0.64	0.73	0.65
CMP1h	0.02355	0.075	0.316	0.054	43 [13,248]	31 [8,218]	0.29	0.19	0.58	0.13	0.73	0.61	0.72	0.66	0.71	0.66
CMP2h	0.03355	0.085	0.323	0.056	49 [15,281]	37 [10,252]	0.33	0.16	0.62	0.11	0.71	0.59	0.70	0.64	0.70	0.64
CMP3h	0.04355	0.095	0.330	0.060	55 [17,314]	43 [12,284]	0.38	0.14	0.65	0.09	0.70	0.57	0.69	0.62	0.68	0.62
CMP4h	0.05355	0.105	0.336	0.062	60 [19,346]	48 [14,315]	0.42	0.11	0.68	0.07	0.68	0.56	0.67	0.60	0.66	0.60
CMP5h	0.06355	0.115	0.343	0.065	65 [21,371]	53 [16,348]	0.47	0.10	0.71	0.06	0.67	0.53	0.66	0.58	0.65	0.58
CMP6h	0.07355	0.125	0.349	0.069	70 [23,399]	59 [18,380]	0.51	0.08	0.74	0.05	0.65	0.50	0.64	0.55	0.64	0.56
CMP7h	0.08355	0.135	0.357	0.073	74 [25,427]	63 [19,410]	0.54	0.07	0.76	0.04	0.64	0.48	0.63	0.54	0.62	0.53

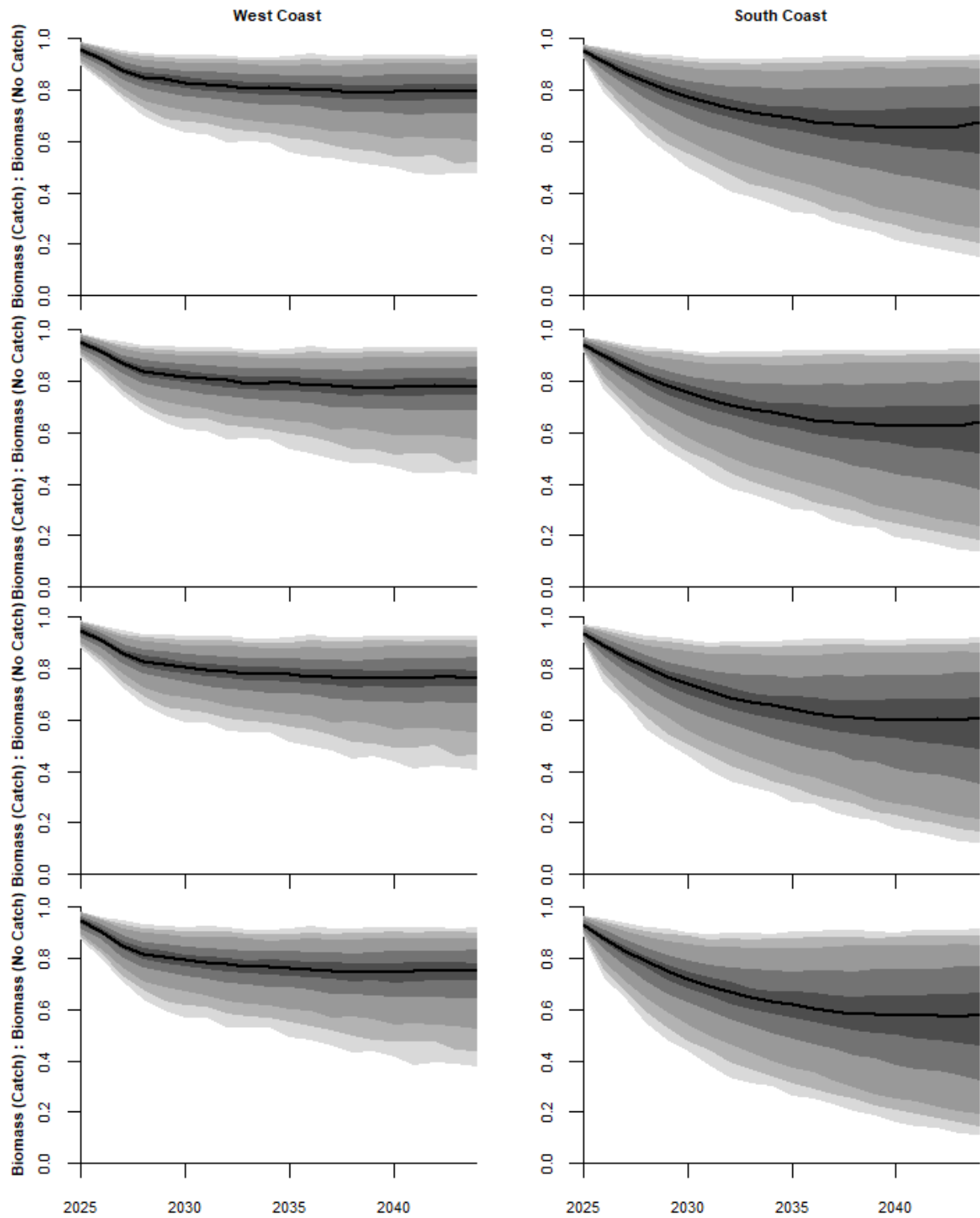


Figure 1a. The median, 20 50, 80, 90 and 95%iles (in increasingly lighter shades of grey) of the annual projected **biomass** under **CMP1 to CMP7** (rows top to bottom) as a proportion of that under a no future catch option for S_{BH} .

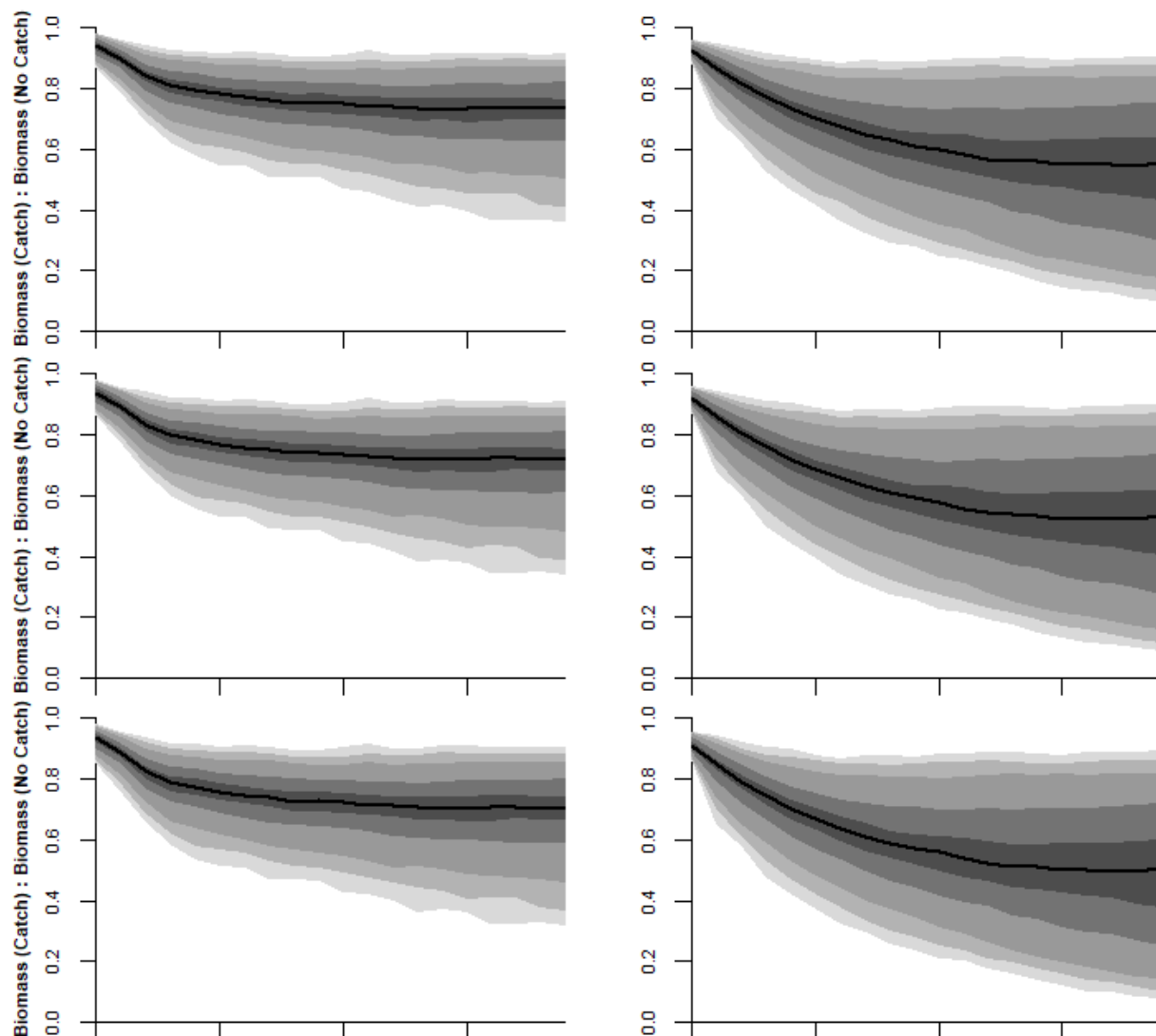


Figure 1a (continued).

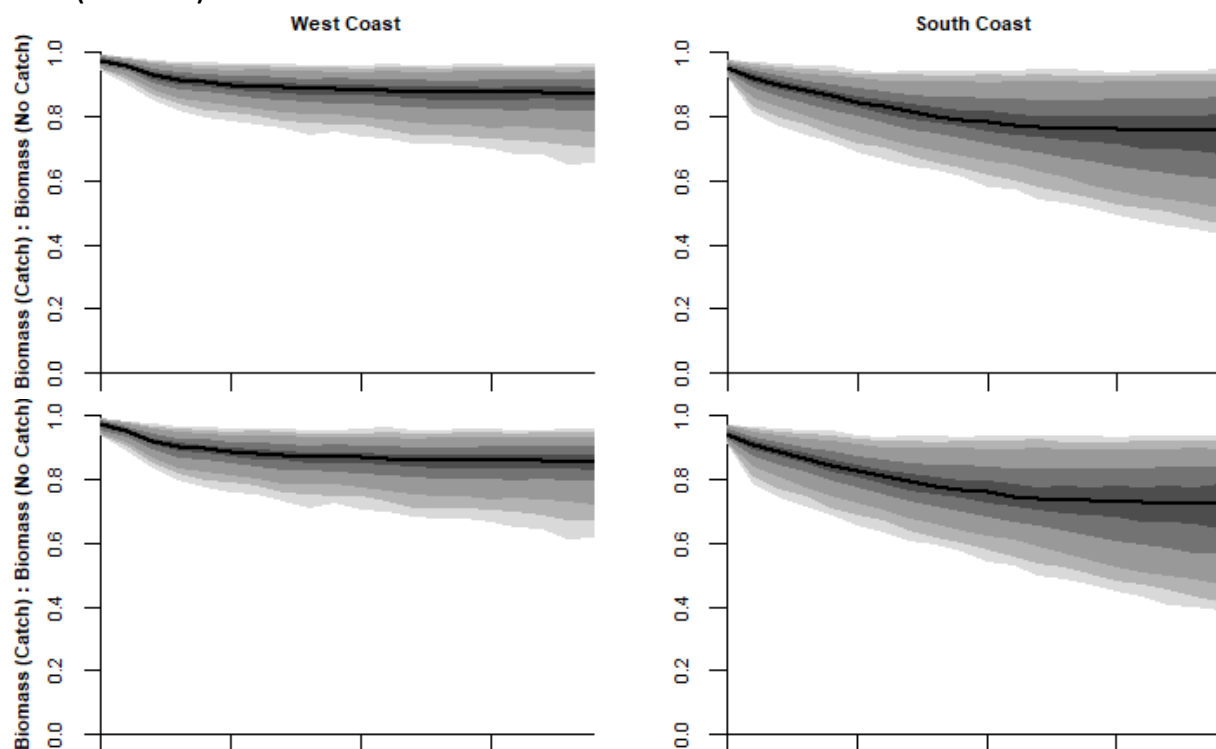


Figure 1b. The median, 20 50, 80, 90 and 95%iles (in increasingly lighter shades of grey) of the annual projected **biomass** under **CMP1b to CMP7b** (rows top to bottom) as a proportion of that under a no future catch option for S_{BH} .

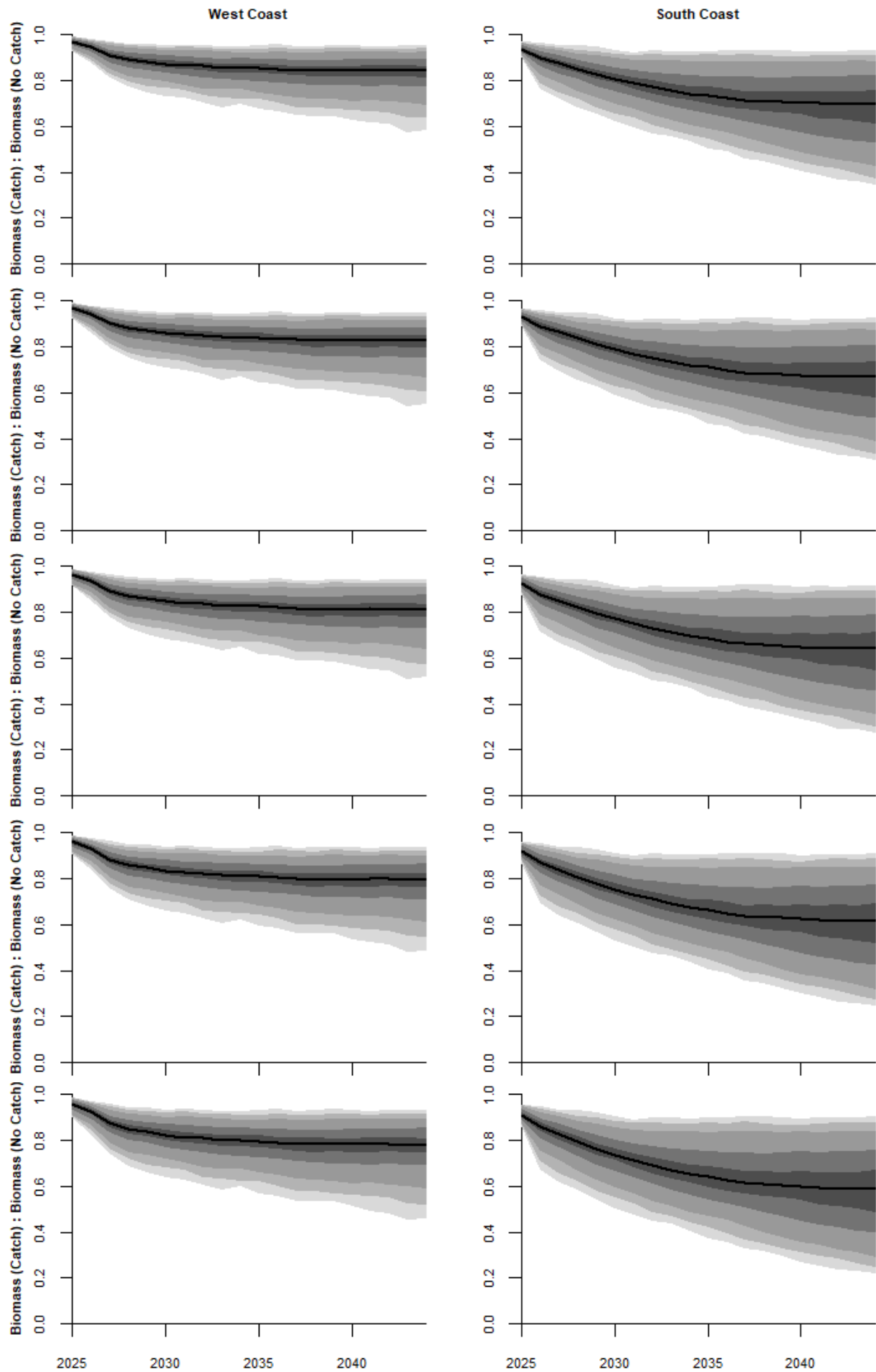


Figure 1b (continued).

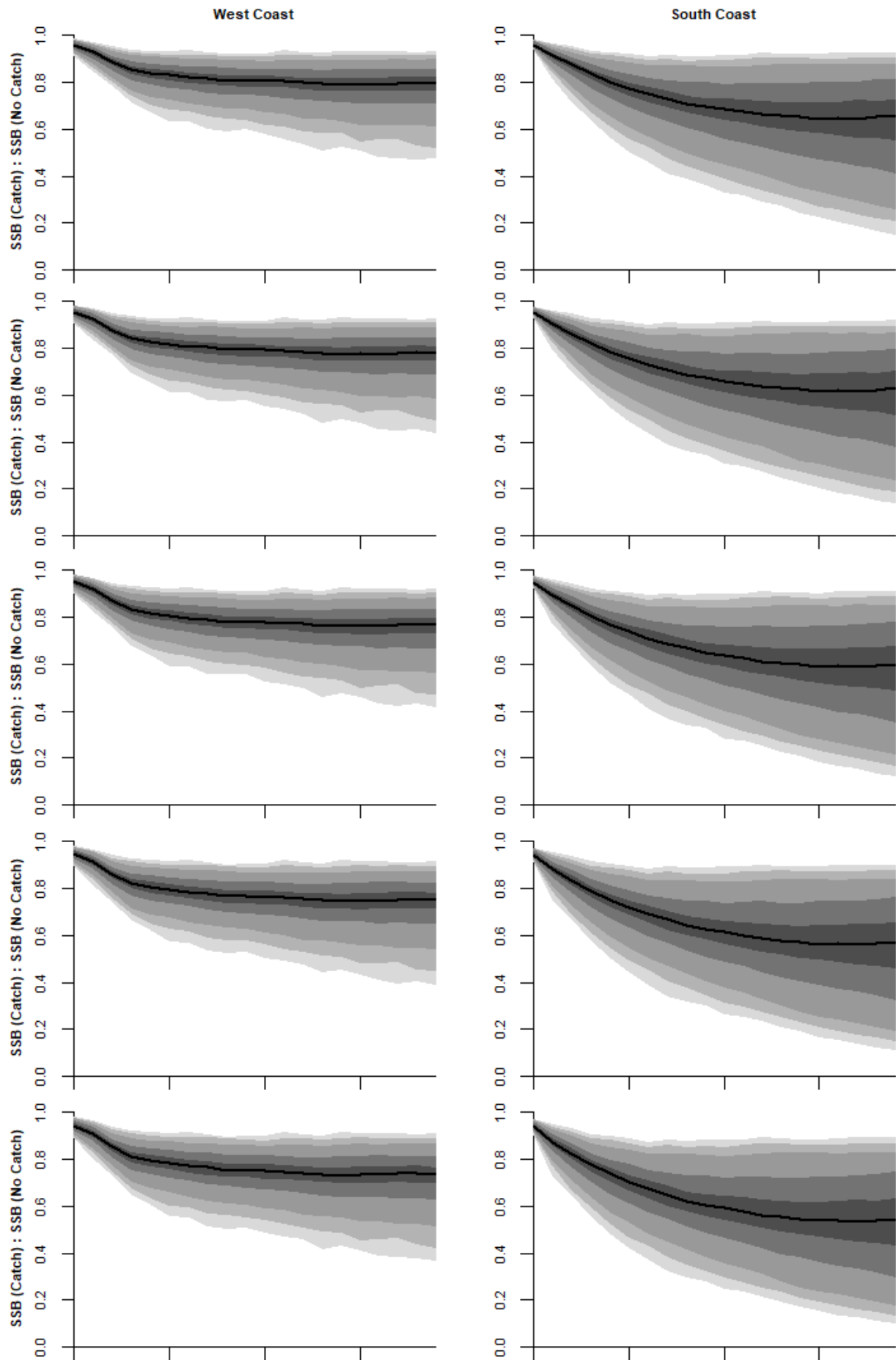


Figure 2. The median, 20 50, 80, 90 and 95%iles (in increasingly lighter shades of grey) of the annual projected **spawner biomass** under **CMP1 to CMP7** (rows top to bottom) as a proportion of that under a no future catch option for S_{BH} .

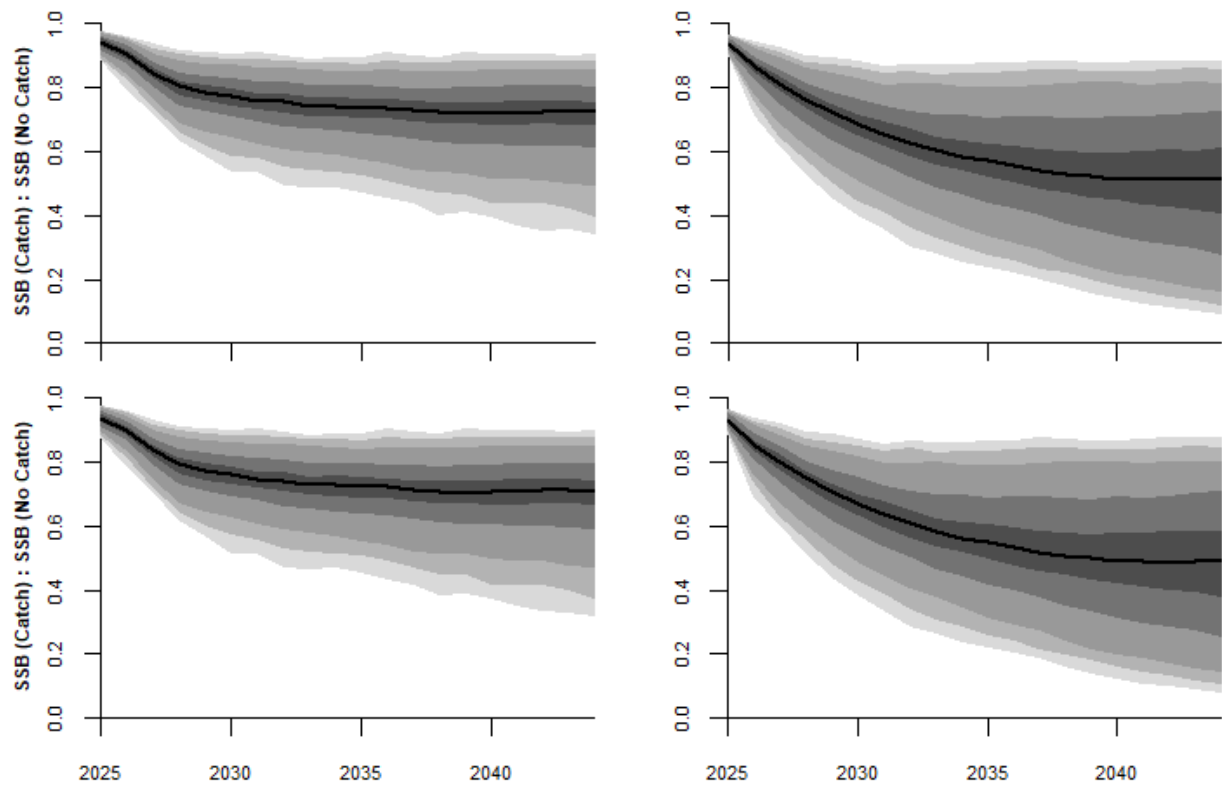


Figure 2 (continued).